

Yogesh Gandhi

PhD, Aerospace Engineering · Finite element simulation, verification and validation, design optimization

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PROFILE

Aerospace engineer who builds finite element capability, proves it is correct, and optimizes with it. Element formulations, buckling and nonlinear stability, modal and dynamic response, material and geometric nonlinearity. Analytical benchmarking and convergence studies that establish whether a result can be trusted. Topology and shape optimization under manufacturing constraints, taken through to manufactured, test validated hardware. First author in *CMAME*.

EXPERIENCE

Senior Researcher · Braid Technologies

Aug 2025 – present

Tokyo, Japan

- Build the differentiable finite element engine underneath Braid's system for going from design intent to manufacturable geometry.
- **Element formulations and verification.** Shell and solid element formulations, with the V&V plans and analytical benchmarks that verify them.
- **Gradient correctness and optimization.** Discrete adjoint and shape sensitivities, verified by Taylor remainder convergence rate and dot product tests, with finite difference step size and Richardson extrapolation analysis. Multi objective and multi constraint shape optimization.
- **Modal dynamics.** Modal superposition, modal transformation and Galerkin projection, mode shapes, per mode stress recovery, modal mass normalization.
- Stack: LS-DYNA, Kratos Multiphysics, MFEM, Python, MATLAB.

Postdoctoral Researcher · University of Bologna

Jan 2024 – Jun 2025

Forlì, Italy

- Design optimization for additively manufactured structural parts, from method to production hardware.
- Stress based design framework for 3D printed Ferrari 296GTB wing mechanism components, measurably lighter and stiffer, validated against test data. Collaboration between UNIBO and Ferrari.
- Co-led design, analysis and 3D printing of parts for the Emilia V solar vehicle, winner of the Sasol 2024 Cruiser Challenge.

Research Associate · University of Parma

Apr 2018 – Feb 2020

Parma, Italy

- Computational composites: shape memory alloys, bistable laminates, delamination and fracture.
- Implemented constitutive models as Abaqus UMAT subroutines in Fortran.

EDUCATION

PhD, Aerospace Engineering · University of Bologna, Forlì, Italy

2021 – 2024

Thesis: a unified geometry projection topology optimization approach for additively manufactured, variable stiffness, continuous fiber reinforced composite laminates.

Visiting collaborations: TU Delft (Prof. Alejandro Aragón), Politecnico di Torino, University of Padova.

MSc, Aerospace Engineering · University of Bologna, Forlì, Italy 2015 – 2017

Final mark 100/110. Thesis: vision based navigation and nonlinear control for an autonomous mobile robot, carried out as a visiting researcher at Coventry University, UK (Apr – Sep 2017). Path planning and control with A*, D* and particle swarm optimization.

BE, Mechanical Engineering · PES Institute of Technology, Bengaluru, India 2010 – 2014

SKILLS

Methods	Structural analysis, nonlinear FEA, buckling and nonlinear stability, material and geometric nonlinearity, modal and dynamic response, shell and solid element formulations, contact mechanics, verification and validation, analytical benchmarking, mesh convergence studies, topology and shape optimization under manufacturing constraints, composite design, additive manufacturing, differentiable simulation, adjoint sensitivities, gradient verification, isogeometric analysis, numerical reproducibility.
Software	Abaqus with UMAT subroutines, LS-DYNA, Kratos Multiphysics, MFEM, GMSH, SolidWorks. ANSYS at familiarity level.
Programming Languages	Python, MATLAB, Fortran. C and C++ at beginner level. English (C1), Italian (B2), Japanese (beginner, learning).

PUBLICATIONS

Around 126 citations, h-index 5, i10-index 4 on Google Scholar.

2025 **Gandhi Y.**, Aragón A., Norato J., Minak G. A geometry projection method for the topology optimization of additively manufactured variable-stiffness composite laminates. *Computer Methods in Applied Mechanics and Engineering*. doi:10.1016/j.cma.2024.117663

2022 **Gandhi Y.**, Minak G. A review on topology optimization strategies for additively manufactured continuous fiber-reinforced composite structures. *Applied Sciences*. doi:10.3390/app122111211

2019 **Gandhi Y.**, Pirondi A., Collini L. Optimal design of shape memory alloy composite under deflection constraint. *Materials*. doi:10.3390/ma12111733

2024 **Gandhi Y.**, Norato J., Pavlovic A., Minak G. A geometry projection method for designing and optimizing additively manufactured variable-stiffness composite laminates. *Acta Polytechnica CTU Proceedings*, 48. doi:10.14311/app.2024.48.0022

2019 Pirondi A., **Gandhi Y.**, Collini L. Strategies for modelling and optimization of bi-stable composite laminates actuated by embedded SMA wires. *Procedia Structural Integrity*, 24. doi:10.1016/j.prostr.2020.02.042

2018 **Gandhi Y.**, Pirondi A., Collini L. Analysis of bistable composite laminate with embedded SMA actuators. *Procedia Structural Integrity*, 12. doi:10.1016/j.prostr.2018.11.075

SELECTED TALKS

- 2nd Int. Conf. Mathematical Modelling in Mechanics and Engineering, Belgrade (Sep 2024)
- 27th Int. Conf. on Composite Structures, Ravenna (Sep 2024)
- 52nd Convegno Nazionale AIAS, Genova (Sep 2023)
- 25th Int. Conf. on Composite Structures, Porto (Jul 2022)

TEACHING AND SERVICE

- PhD and MSc co-supervision, University of Bologna.
- Topology optimization lecturing: MSc Mechanical Engineering, University of Bologna; MSc Automotive Engineering, Dallara Academy.
- Peer reviewer for *Composite Structures* and *Additive Manufacturing*.